

Group-level medians with 95% bootstrap intervals

Method	Rq error (nm) ↓	PSD distance ↓	Descriptor MAE ↓	SSIM ↑	Diversity ratio
Unconditional train mean	1.000 [0.089, 7.229]	0.847 [0.451, 1.060]	1.471 [0.265, 2.170]	0.038 [-0.005, 0.190]	0.000 [0.000, 0.000]
Nearest-RHEED retrieval	5.775 [0.056, 8.450]	0.424 [0.167, 1.067]	1.708 [0.846, 2.888]	0.018 [0.009, 0.048]	0.000 [0.000, 0.000]
RHEED-conditioned CVAE	0.822 [0.463, 6.763]	0.396 [0.173, 1.014]	2.713 [2.388, 3.847]	0.007 [-0.083, 0.149]	0.676 [0.469, 0.777]